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ABSTRACT

Magnetic mirror configurations are observed in natural settings and have various applications in laboratory plasmas, such as a magnetic expander of the open mirror fusion devices. The axial plasma flow in open mirror systems is significantly influenced by atomic processes involving neutrals, such as ionization and charge-exchange collisions. A quasi-two-dimensional computational model was developed to study these effects on accelerated plasma flow in magnetic mirror configurations. This model includes an emitting wall, a quasineutral flow/acceleration region with a magnetic expander, and a recycling/absorbing wall. Implemented in a hybrid quasineutral code, the model incorporates drift-kinetic ions, fluid electrons, and fully kinetic neutral atoms with collision processes simulated using the direct simulation Monte Carlo approach. Ion recycling on the wall is accounted for using empirical methods. The model demonstrates that slow atoms with short mean free paths create a dense plasma layer near the wall, modifying the plasma potential which can lead to large-scale perturbations due to ion–ion streaming instabilities.

I. INTRODUCTION

Large linear fusion devices use the magnetic expander (diverging magnetic field configuration) to distribute the output heat loads over the larger area.[1] The divertor with large expansion can also reduce electron energy losses.[2] The schematic of a linear fusion device with a divertor and the wall is shown in Fig. 1, where the "source" core represents some plasma confinement configuration, e.g., field-reverse configuration.[3] Hot plasma from the source is accelerated by the electric field through the mirror throat toward the wall while the same electric field confines electrons. This study aims to reproduce the dynamical plasma flow effects in the magnetic expander on the time scales of the ion transit motion neglecting plasma rotation but focusing on the effects associated with neutral atoms and collisions. Charge exchange (CX) and ionization create additional ion sources in the near-wall region that decrease the global electrostatic potential drop and may lead to turbulence caused by counterstreaming ions. Such effects may increase electron heat losses due to the lower potential barrier entrapping electrons inside the core, thus resulting in the cooling of the plasma core.[2]

Generally, full electron and ion dynamics require kinetic treatment and should be considered with multidimensional full kinetic simulations, e.g., such as in Refs. 4–6. Such simulations, while could be comprehensive, demand large computational resources and also could be difficult for interpretation. Simplified reduced models are attractive and often offer good insights on key physical phenomena. With the focus on the effects of ion interactions with neutrals, we neglect the electron kinetics and consider only ambipolar ion acceleration in a quasineutral plasma. Models solving both ion and electron motion including the non-quasineutral effects and electron collisions were considered in Refs. 6–8. Plasma flow in the magnetic mirror is essentially (at least) a two-dimensional problem.[9,10] However, a simplified reduced treatment is possible where some two-dimensional effects are modeled by effective losses/sources imitating particle losses due to the radial motion.[11] Another approach is based on the drift-kinetic approximation for the particle motion resolving the one-dimensional dynamics along the magnetic field line. This approach assumes strong magnetization (small Larmor radius) and takes into account the mirror

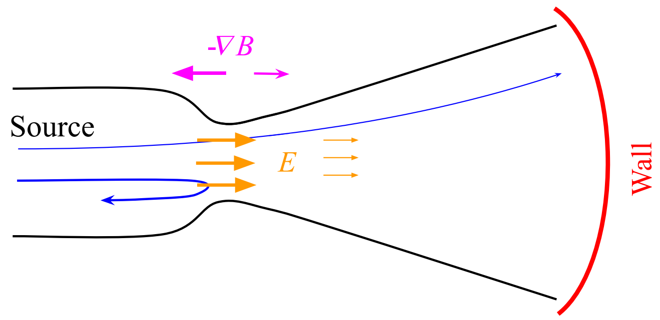

FIG. 1. Schematic of the divertor configuration for a linear fusion device. Plasma escaping the throat of the mirror expands and reaches the wall. Magnetic mirror force confines the source particles and accelerates those escaping into the expander. Quasineutral plasma forms the electrostatic potential (shown in orange), increasing ion losses and confining electrons. Ions showed schematically with blue trajectories.

force.[7,8,12] In the paraxial approximation,[7,8,12] the coordinate along the magnetic field line can be approximated as an axial coordinate and, effectively, the problem can be considered one-dimensional with the magnetic field being a function of the axial coordinate. Physically, this corresponds to the magnetic flux tube with a given flux, and all variables are averaged over the transverse cross section. In this approach, plasma flows along the magnetic field lines. Plasma radial transport, which can occur due to the ion inertial effects and turbulent (azimuthal) fluctuations,[13] is neglected here.

Here, we use the hybrid model that includes drift-kinetic ions in the paraxial approximation, kinetic neutral atoms, and fluid (Boltzmann) electrons. The two-dimensional spreading of unmagnetized neutrals is modeled by the effective loss terms, which depends on the distance from recycling wall. In our paraxial model, and with assumption of Boltzmann electron distribution, the radial plasma gradients and related diamagnetic effects[14] are neglected.

Ions bombarding the material wall create several groups of neutral atoms of various energies. First, ions can be reflected via the backscattering from the material surface, returning predominantly to the neutral state.[15] Such reflected neutrals are of relatively high energy (which is some fraction of the energy of incident ions), and thus, we will call them energetic neutrals. Then, the cold atoms and molecules are emitted via the desorption process. Desorbed atoms are of the wall temperature and, thus, constitute the cold component. Desorbed molecules may undergo ionization and further dissociation creating the so-called Franck–Condon (FC) atoms, of 1–4 eV energy (about the energy of dissociation). In the energy range, Franck–Condon atoms stand between cold atoms (of wall temperature) and energetic atoms due to reflection. Cold atoms have a larger tendency to accumulate in the near-wall region. The CX and ionization can effectively create a source of cold ions and electrons near the wall flowing in the opposite direction (from the wall toward the plasma core). Plasma–neutral interactions were considered previously by using hydrodynamic type balance equations,[16] and specifically in the magnetic mirror configuration in Ref. 17. However, in many cases, the mean free path for neutrals is large and neutrals have to be considered kinetically.[18] Here, we use a kinetic model to describe plasma interactions with neutrals at different energies and also include the geometry effects to describe the natural depletion of the neutrals back-flow flux due to the three-dimensional spreading as they move away from the recycling wall. It is shown here that a sufficiently dense cold ion beam generated in the direction opposite to the original ion flow strongly affects the potential distribution near the wall and may trigger an instability that results in turbulent behavior near the wall. As for the energetic neutral atoms, they quickly leave the system without significant effects on the plasma flow.

In this paper, we use a semi-empirical model of the atoms yield from the recycling wall to study their effects on the plasma flow. The model includes an emitting source with a diffusive or specular reflection for ions reflected back to the emitting wall, an atom recycling model for the ions reaching the material wall, producing various groups of atoms, and the most probable collision processes such as CX and ionization. Heavy particles (ions and neutrals) are modeled with the particle-in-cell (PIC) method. The collision processes are simulated with the direct simulation Monte Carlo (DSMC). The ion–neutral CX collisions depend on the corresponding particle distribution functions and the reaction cross sections. The constant collision rate is assumed for the ionization process due to isotropic electrons. For a large part of the simulated domain, mean free paths for collision processes are on the scale of system size and higher; thus, the DSMC method is justified in this regime. Note that background atom density is neglected and neutral atoms presented in the model are only due to plasma–wall interaction model.

The paper is organized as follows. In Sec. II, we provide a general description of the quasineutral drift-kinetic model, with more details on the atom transport in Sec. II A, and collisions in Sec. II B. The results are presented in Sec. III, demonstrating the plasma dynamics in the absence of collisions, Sec. III A; the effect of core's ion temperature, Sec. III B; and the ion–ion streaming instability induced by the presence of cold ion source near the wall, Sec. III C.

II. HYBRID DRIFT-KINETIC MODEL

Plasma flow in the axisymmetric converging–diverging mirror configuration is modeled as a one-dimensional flux tube in the paraxial approximation, where the radial component of the magnetic field B_r is much smaller than the axial B_z. Below we assume the total magnitude of the magnetic field $B \simeq B_z$. With the variable plasma cross section, this model accounts for two-dimensional effects related to compression and decompression of the plasma flux tube near the axis of the magnetic configuration. Electrons are taken in the Boltzmann approximation with full plasma quasineutrality over the whole domain (near-wall sheath effects are ignored), as we focus on the low-frequency plasma dynamics. Particles drift in the axisymmetric magnetic configurations, i.e., the rotation in the azimuthal direction, which may cause azimuthal currents and diamagnetic effects are neglected.

Ions are modeled in the drift-kinetic approximation, i.e., following the gyrocenters on time scales larger than the gyro-frequency, and on spatial scales larger than the ion gyroradius (neglecting gyroradius effects). The timescale of interest here is the transit time in the large magnetic mirror system, $\tau \approx L/v_z$ (v_z is ion velocity along magnetic field line), e.g., for the hydrogen ion with energy of $\sim 100\,\mathrm{eV}$ and the system length $L \approx 1\,\mathrm{m}$, $\tau \approx 10\,\mu\mathrm{s}$. We assume that the magnetic moment is conserved for ions, i.e., $\rho_i < L_B \equiv (|\nabla B|/B)^{-1}$ (magnetic field is nearly homogeneous per single ion rotation). With the magnetic moment conservation, ion gyrocenters experience the electric field force and the mirror force

$$m_i \frac{dv_z}{dt} = eE_z - \mu \frac{\partial B}{\partial z}, \tag{1}$$

where the magnetic moment $\mu = m_i v_\perp^2/2B = \text{const}$. The ion distribution function (IDF) f_i follows the Boltzmann equation:

$$\frac{\partial f_i}{\partial t} + v_z \frac{\partial f_i}{\partial z} + \frac{dv_z}{dt}\frac{\partial f_i}{\partial v_z} = \mathcal{C}_{i,a}(f_i, f_a), \tag{2}$$

where $\mathcal{C}_{i,a}$ is the collision operator and f_a is the atom distribution function (ADF). The collision operator $\mathcal{C}_{i,a}$ is accounted with the DSMC algorithm (more details in Sec. II B).

In this model plasma, source is modeled by the emitting wall at the left boundary. Ions are injected from the left (source) wall with the flux Maxwellian distribution parallel along the z-direction and isotropic Maxwellian for the perpendicular component

$$f_i(v_z, v_\perp) = \frac{2n_0}{\pi^{3/2} V_{Tz} V_{T\perp}^2} \exp\left(-\frac{v_z^2}{V_{Tz}^2}\right) \exp\left(-\frac{v_\perp^2}{V_{T\perp}^2}\right), \tag{3}$$

for $v_z > 0$, injecting the flux-Maxwellian distribution $v_z f_i(v_z, v_\perp)$ using standard loading techniques.[19] The constant factor of "2" in Eq. (3) ensures $n_0 = \int f_i dv_z 2\pi v_\perp dv_\perp$ since we inject in the half-space. Here, v_{Tz} and $v_{T\perp}$ are parallel and perpendicular thermal velocities, respectively, defined as $v_T = \sqrt{2T/m}$ (m is the ion mass), where the corresponding temperatures are defined as

$$T_z = \frac{2}{n_0} \int m v_z'^2 \hat{f}_i dv_z 2\pi v_\perp dv_\perp, \tag{4}$$

$$T_\perp = \frac{1}{2n_0} \int m v_\perp^2 \hat{f}_i dv_z 2\pi v_\perp dv_\perp, \tag{5}$$

where $v_z' = v_z - V_{0z}$ and $V_{0z} = V_{Tz}/\sqrt{\pi}$ is the average injected flow velocity, $\hat{f}_i = f/n_0$ is the normalized distribution function. Some ions may be reflected back by the magnetic mirror force (or some fluctuations in the electric field) and reach the source wall. We consider a fully absorption boundary condition for the source wall such that all ions returning to the wall are removed.

During the relatively slow ion bounce motion in the magnetic mirror, electrons quickly fill the electron loss cone due to Coulomb collisions and cause higher electron losses. Plasma quasineutrality gives rise to the electrostatic potential slowing electron losses but expands the ion loss region in the velocity space.[20] It allows to consider Maxwellian and isotropic electrons and, in the simplest case, use the Boltzmann relation for electrons,

$$0 = -enE - T_e \nabla_z n, \tag{6}$$

where the full quasineutrality is assumed, $n = n_e = n_i$. In this approximation, the density gradient generates the electric field which accelerates ions.

Each time iteration consists of the ions injection, interpolating plasma density on the grid, solving for the electrostatic potential via the Boltzmann relation, and integrating particle motions, Fig. 2. When the ions reach the material wall, neutral atoms are emitted in accordance with the atom transport model (see Sec. II A). Collisions are processed at the constant time step Δt_{coll} (which is equal or larger than the simulation time step), and the collisional cells coincide with the PIC cells of size Δx.

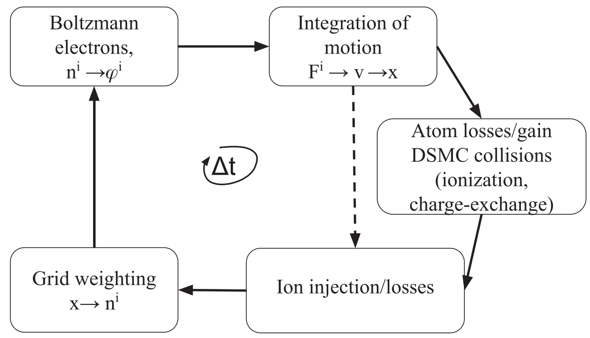

FIG. 2. Schematic representation of the integration time step.

The collision processes satisfy $\Delta x/\lambda_{mfp} \ll 1$ and $\Delta t_{coll}\nu_{coll} \ll 1$, where λ_{mfp} and ν_{coll} are average mean free path and average collision frequency, respectively.[21] The schematic of material wall transport and collision processes is shown in Fig. 3 and discussed in detail below.

A. Neutral dynamics

The reflection of atoms bombarding the wall is one of the main processes occurring during plasma wall bombardment. Light atoms hitting a wall are scattered within the material surface (after one or multiple collisions) and at a large enough angle can be effectively reflected. Note that a portion of reflected particles can remain charged as shown in Ref. 22, which is neglected in this model, assuming all reflected particles are electrically neutral. In addition to the reflection process, we include two types of desorbed atoms: cold atoms of about wall temperature, and more energetic atoms created due to molecular desorption and its further ionization and dissociation, the so-called Franck–Condon atoms with energies $1-4$ eV. The latter are included in our model empirically via a parameter without an actual model of the molecular dissociation. Due to the general difficulties for accurate accounting of complex

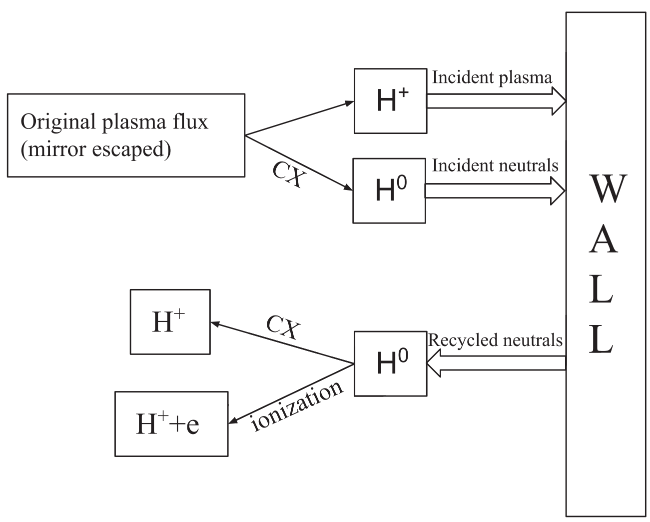

FIG. 3. General schematic of atom transport with modeled collision processes. Ions, accelerated to supersonic velocities through the magnetic throat, enter the near-wall region populated with atoms and hit the wall.

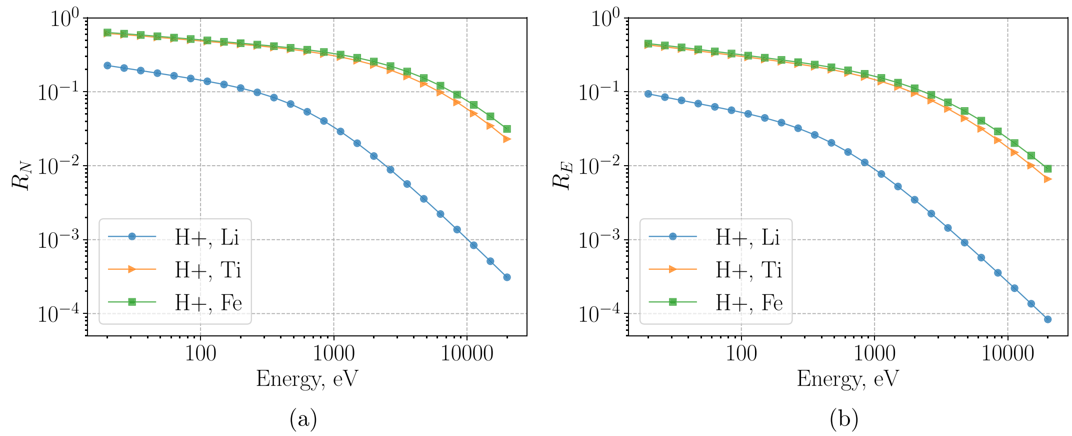

FIG. 4. Total reflection coefficients (a) R_N and (b) R_E as a function of normal incident energy for hydrogen ion bombarding material wall (Li, Ti, or Fe), obtained from Eq. (7) with appropriate coefficients are taken from Ref. 15.

chemistry processes on the wall, the desorption process is modeled empirically and controlled by the sticking coefficient and other parameters. Below, more details are provided on the modeling of reflection desorption processes, which are included in the model.

1. Reflection model

The reflection coefficients are calculated from both experiments and simulations for a wide range of incident particles and materials.[15] They are mainly given as integral yield coefficients for the fraction of reflected particles R_N and the fraction of reflected energy R_E relative to the corresponding number and energy of incident particles. Higher-dimensional information, such as the distribution function of the reflected population and its dependence on the incident energy and the incident angle, is only available for limited situations. Experimentally validated computer simulations are also not widely available.

The relation between R_N and R_E only contains information on average reflected energy. Coefficients R_N, R_E are functions of the incident energy E_0 and incident angle α (typically evaluated at normal incidence angle $\alpha = 0°$). In our setup, the perpendicular pressure of incident particles is low (in the end of the expander region), and the normal incidence $\alpha = 0°$ is assumed.

The semi-empirical formula was proposed for both R_N, R_E that can be used for a wide range of materials:[15]

$$R_N\ (\text{or}\ R_E) = \frac{A_1 \ln A_2\epsilon + e}{1 + A_3\epsilon^{A_4} + A_5\epsilon^{A_6}}, \tag{7}$$

where $A_1 - A_6$ are the fitting parameters (they differ for R_N and R_E), e is the base of natural logarithms, and ϵ is the Tomas–Fermi reduced energy (sometimes Firsov screening length is used, giving $\sim$10% smaller ϵ[22]). The reduced energy ϵ is the function of the incident particle energy E_0, its mass ratio with target material nuclei μ, and electric charges of both the incident particle and the target nuclei (its atomic number).[15] The coefficients R_N, R_E are depicted in Figs. 4(a) and 4(b) as a function of incident energy.

Alternatively to R_E coefficient, one can utilize the energy distribution function $F_E(E_0, \theta)$ of the reflected particles integrated over the reflection azimuthal angle, where E_0 is the incident energy and θ is the incident polar angle. The empirical relation for F_E is proposed in Ref. 23 in the form

$$F_E = A\left[\left(C + D(E/E_0)^2\right)(B - E/E_0) + \exp\left(-(E/E_0 - B)^2/2\xi^2\right)\right], \tag{8}$$

where B, C, D, ξ are the fitting parameters, available for the following pairs: (H^+,Cu), (He^+,Cu), (He^+,Ni), for the incident energies 1 keV, 5 keV, and the incident angles of $(0°, 40°, 60°, 70°, 80°)$.[23] The coefficient A in Eq. (8) is found via

$$\int_0^{E_0} F_E dE = N_0 R_N, \tag{9}$$

where N_0 is the number of incident particles, and R_N can be evaluated from Eq. (7). The example distributions F_E produced for the three different incident energies are depicted in Fig. 5, for the (H^+,Cu), normal incidence, and incident energy of 1 keV.

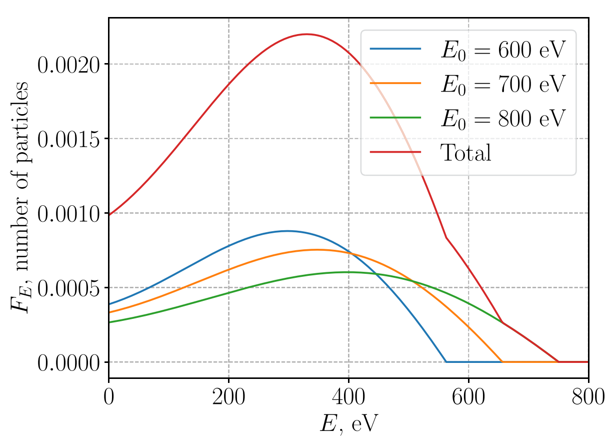

FIG. 5. Examples of the reflected energy distribution function F_E for neutrals according to model in Ref. 23, for three incident particles with energies 600, 700, 800 eV, where coefficients in Eq. (8) are for (H^+,Cu) (hydrogen ions bombards Cu wall); incident particle number $N_0 = 1$.

The polar angle for reflected particles essentially defines the ratio of parallel and perpendicular energy of the reflected yield. In our model, the azimuthal angle for reflected particles is not resolved, assuming the isotropic distribution for the azimuthal angle. Both experiments and simulations suggest that the cosine distribution is a good approximation for the polar angle distribution for the incident energies below $\epsilon \approx 10$, according to Ref. 22. For example, for the pair (H^+,Ti), this range translates to incident energies below 20 keV which suffice our needs. Thus, we proceed by sampling the energy from the distribution F_E using Eq. (8) and then sampling the polar angle for every particle from the cosine distribution.

In all further simulations, we utilize R_N coefficients for the reflected yield, Eq. (7), which are evaluated for hydrogen ions bombarding the titanium wall. As for the energy distribution function F_E, it is taken according to the empirical formula (8) with the coefficients corresponding to the pair (H^+,Cu), normal incidence, and 1 keV incident energy.

2. Desorption model

In addition to high-energy reflected particles, low-energy atoms also "peel off" the wall due to desorption. These atoms have a low velocity and can accumulate in the region close to the wall, which can significantly affect overall plasma dynamics. To model the desorption process, a more qualitative and empirical approach will be taken, unlike the reflection model described earlier. The desorbed atoms are assumed to come from two sources: wall-temperature (cold) atoms and Franck–Condon (FC) atoms, with energies ranging from 1 to 4 eV, which are close to the dissociation energy. In the equilibrium, the following relationship is assumed between the reflection coefficients:

$$1 = P_r + P_d + P_a, \tag{10}$$

where P_r is the reflection ratio, namely, $P_r \equiv R_N$; P_d is the total desorption ratio; and P_a is the adsorption ratio (also called sticking coefficient). We assign the constant isotropic temperatures for low-energy (cold) and FC atoms, $T_{a,c}$ and $T_{a,FC}$, respectively. The desorbed yield P_d further split with the coefficient P_{FC} (0–1), defining the probability of injecting a Franck–Condon atom by using the rejection sampling. A random number $0 \le r \le 1$ is sampled: if $r \le P_{FC}$, then the injected atom is assigned the temperature $T_{a,FC}$, otherwise $T_{a,c}$. For the total number of recycled particles (reflected and desorbed), Eq. (10), the parameter $\mathcal{R} = P_d + P_r$ is introduced similar to the DEGAS neutral model,[24] effectively replacing P_a coefficient, $P_a = 1 - \mathcal{R}$.

3. Radial losses

Atoms from all sources described above can escape the simulation volume in the radial direction (which is not resolved, but the particle's perpendicular velocity allows to estimate its radial dynamics). We add this effect by assigning the virtual radial coordinate to every atom, with the initial value $r = 0$ cm. Whenever a particle crosses the radial length L_r, it is removed. First and foremost, the radial losses cause the significant drop of atom density with the distance from the wall, as illustrated by the resulting atom densities profiles in equilibrium in Fig. 6 for the expander region (the region between the mirror throat and the wall). More detailed phase space plots of atom distribution function (ADF) with the radial losses effects were presented before.[25] Obviously, the losses significantly increase when the $L_r \ll L$

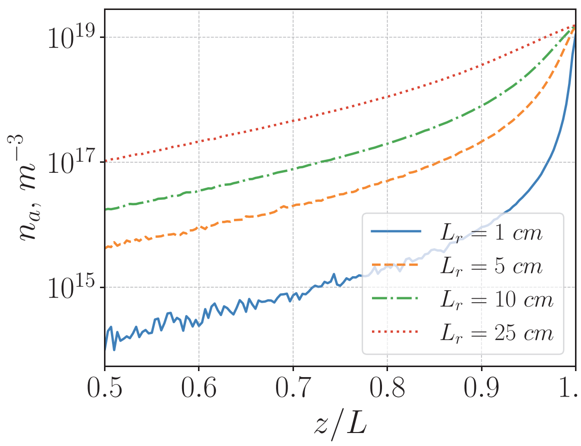

FIG. 6. Atom density profiles for the various radial loss coefficients L_r (radial "wall"), demonstrating pure radial loss effect (without collisions).

(for isotropic atom population), where L is the system size in the axial direction. Parameters of the test simulations are given in Table I, and the radial loss coefficient is varied, $L_r = (1, 5, 10, 25)$ cm.

B. Collisions with DSMC

Most probable collision events for the parameters of interest are the ion–neutral CX and ionization of neutrals due to electron–neutral collisions. Direct simulation Monte Carlo (DSMC) algorithm is implemented for handling collisions with the null collision method.[21,26] Since the ions and neutrals are both kinetic, CX collisions are calculated self-consistently with their corresponding distribution functions. For the ionization, since electrons assumed isotropic, $T_e = \text{const}$, the constant value of the ionization rate is used, $\beta_i \equiv \langle \sigma v \rangle = 2.1 \times 10^{-14}\ \text{m}^3\ \text{s}^{-1}$ (averaged over Maxwellian distribution for $T_e = 200$ eV),[27] where σ is the collisional cross section and v is the relative speed of interacting particles. Notation $\langle . \rangle$ here represents the statistically averaged quantity of the collision rate over a large number of events. The cross sections for CX ion–neutral collisions, (H^+, H) $\rightarrow$ (H, H^+), are obtained from Ref. 27, reaction 3.1.8.

To minimize errors associated with the DSMC technique, we kept the cell size much smaller than the mean free path, $\Delta x \ll \lambda$, and well resolve the collision frequency $\nu \Delta t_{coll} \ll 1$, Ref. 21. The subcycling integer parameter $N_{sub,c}$ is introduced for collisional time step, relating it to the simulation time step as $\Delta t_{coll} = N_{sub,c} \Delta t$. Additionally, a particle tracking data structure was used to minimize the memory consumption, which was described earlier in Ref. 25.

III. RESULTS

Since the recycling model has many empirical parameters, we focus on establishing the main effects and discuss its mechanisms. Hydrogen mass is taken for ion and neutral species, and only singly

TABLE I. Simulation parameters for the case demonstrating radial loss effects.

L (m)	$\Gamma_0{}'$ ($\text{m}^{-2}\ \text{s}^{-1}$)	T_e (eV)	T_i (eV)	$\mathcal{R}$	P_{FC}	$T_{a,c}$ (eV)	$T_{a,FC}$ (eV)
3.0	3×10^{23}	200	50	0.95	0.5	0.1	2.0

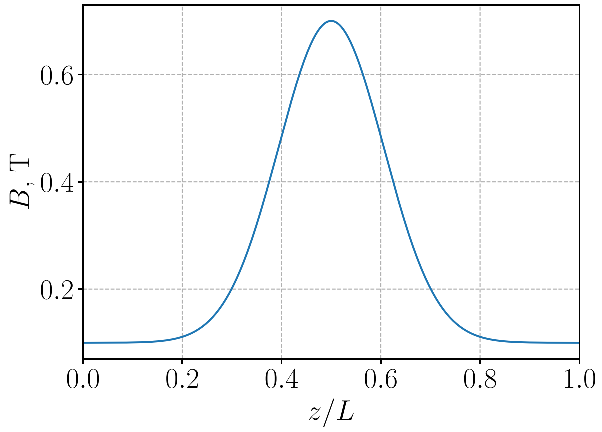

FIG. 7. Converging–diverging magnetic field profile given by Eq. (11) with the magnetic throat at the center, with mirror ratio $R = 7$.

charged ions are assumed. In all of the following simulations, the magnetic field intensity is given by

$$B = a + b \exp\left(-\frac{(z - z_0)^2}{\delta_B^2 L^2}\right), \tag{11}$$

where $a = 0.1\,\mathrm{T}$, $b = 0.6\,\mathrm{T}$, $z_0 = L/2$, $\delta_B = 0.15$, hence the mirror ratio $R = 7$. The profile of the magnetic field is shown in Fig. 7.

First, plasma acceleration without atoms is shown to demonstrate the basics physics of the ion acceleration in the mirror. The chosen parameters are as follows: the system length $L = 3\,\mathrm{m}$, a uniform electron temperature $T_e = 200\,\mathrm{eV}$, injected (half-Maxwellian) ion temperature $T_i = 100\,\mathrm{eV}$, injected flux $\Gamma_0 = 10^{23}\,\mathrm{m}^{-2}\,\mathrm{s}^{-1}$. The resulting density, potential, and velocity profiles of the quasineutral plasma flow are shown in Figs. 8(a)–8(c). Ion (plasma) density shown in Fig. 8(a) is normalized to the injected density

$$n_0 = \Gamma_0 / v_0, \tag{12}$$

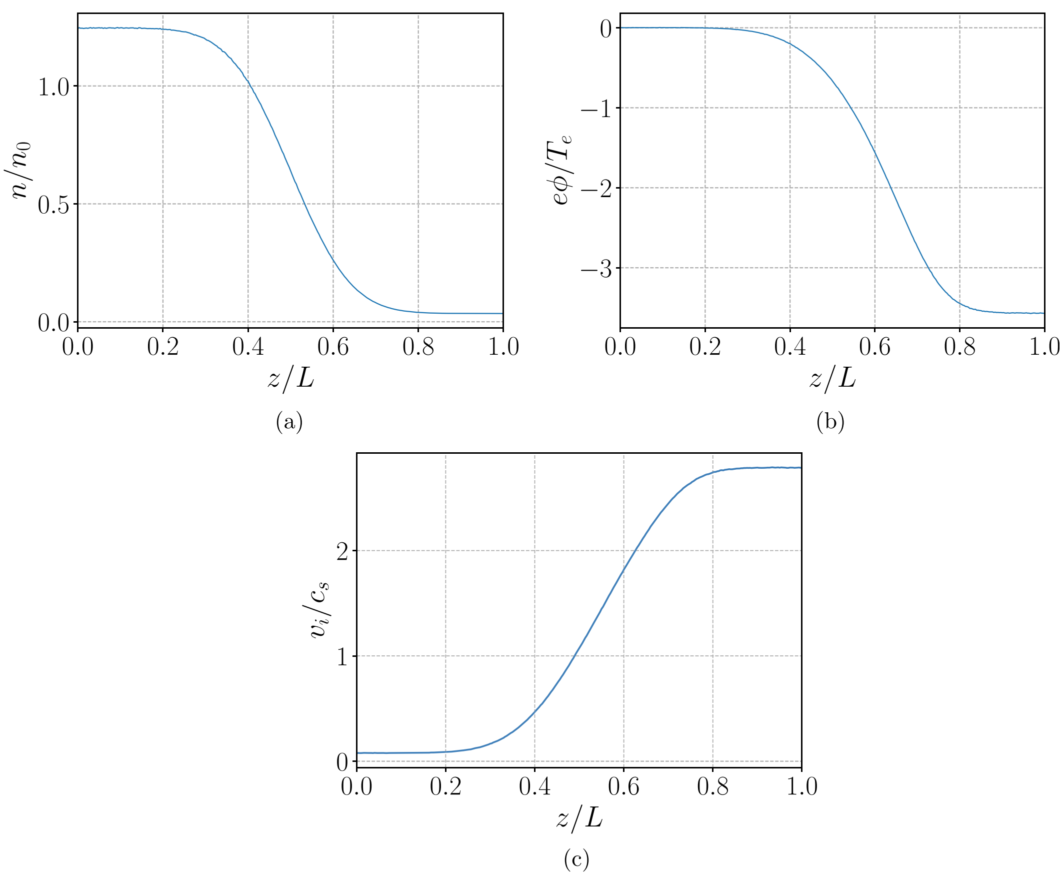

FIG. 8. (a) Spatial profiles of plasma density, (b) electrostatic distribution, and (c) ion velocity in whole domain for the case without atoms, with pure plasma acceleration. Note the transition through ion-sound velocity at around the maximum of magnetic field (at the center); it is not precise ($v_i/c_s = 1$) due to the definition $c_s^2 \equiv T_e/m_i$ which neglects the ion temperature effect.

TABLE II. Simulation parameters for Base case with collisions.

$L\,(\mathrm{m})$	$\Gamma_0{}'\,(\mathrm{m}^{-2}\,\mathrm{s}^{-1})$	$T_e\,(\mathrm{eV})$	$T_i\,(\mathrm{eV})$	$\mathcal{R}$	P_{FC}	$T_{a,c}\,(\mathrm{eV})$	$T_{a,FC}\,(\mathrm{eV})$
3.0	3×10^{23}	200	100	0.95	0.5	0.1	2.0

where $v_0 = v_{Ti}/\sqrt{\pi}$ is the average flow velocity for the half-Maxwellian distribution, Eq. (3). The value on the left boundary in the range 1–2 indicates the fraction of reflected ions. The established total electrostatic potential drop is about 3.7 electron temperatures (i.e., $e\Delta\phi/T_e$) which is ~ 700 V, Fig. 8(b). The electrostatic potential follows the electron Boltzmann distribution, Eq. (6). As will be shown in the simulations with neutral transport, plasma density will increase in the expander region (mainly close to the wall) as the additional plasma sources are introduced via ionization and CX. This will lead to the overall drop in the potential, which may be interpreted as poor electron confinement in the source region.

Below we present the base case with collisions, defining the main sources of particles (both ions and neutrals) near the wall and their main effects. After that, we demonstrate how various parameters affect the overall solution, such as core ion temperature, and the fraction of desorbed atoms (cold vs Franck–Condon).

A. Base case with collisions

Here, in addition to the full atom transport model (with radial losses), we include collisions due to the CX and the ionization process. The full list of parameters for this case is given in Table II. This case will be referred to as Base case in further numerical experiments.

The resulting macroscopic quantities for this case presented in Figs. 9(a)–9(c). Note, the atom flow velocity increases in the opposite direction to the ion flow, as the faster atom component has smaller probability to be involved in a collision process (atom mean free path depends on the atom flow velocity).[28] The global electrostatic potential drop $\Delta\phi$ decreased about two times compared to the case without atom effects due to additional plasma sources present near the wall.

The mean free path for atoms is smaller than that for ions due to the low average velocity of atoms and higher collision frequency. We define the mean free path for atoms as

$$\lambda_a = \frac{v_a}{n\left(\langle\sigma v\rangle_{cx} + \langle\sigma v\rangle_{ion}\right)}, \tag{13}$$

where v_a is the atom flow velocity and $\langle\sigma v\rangle$ denotes the average collision frequency in a cell (CX or ionization). The collision rate for CX events is obtained as $\beta_{CX} = \langle\sigma v\rangle_{cx} = S_{CX}/n_a n$, where $S\,(\mathrm{m}^{-3}\,\mathrm{s}^{-1})$ is

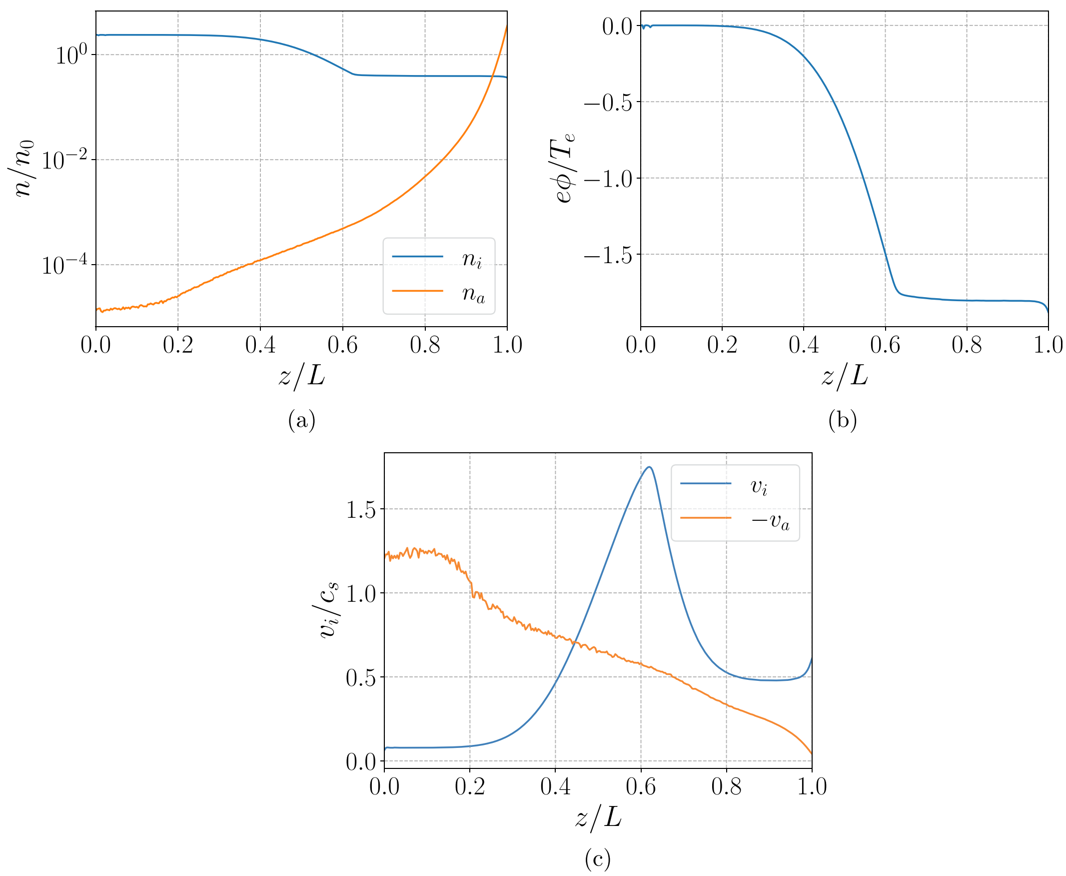

FIG. 9. (a) Spatial profiles of plasma and atom density, (b) electrostatic potential, and (c) flow velocities of ions and atoms for Base case. Note that the ion flow velocity profile sagged substantially due to low energy ion source generated in the near-wall region. Also, the global electrostatic potential dropped about two times due to increased plasma density near the wall.

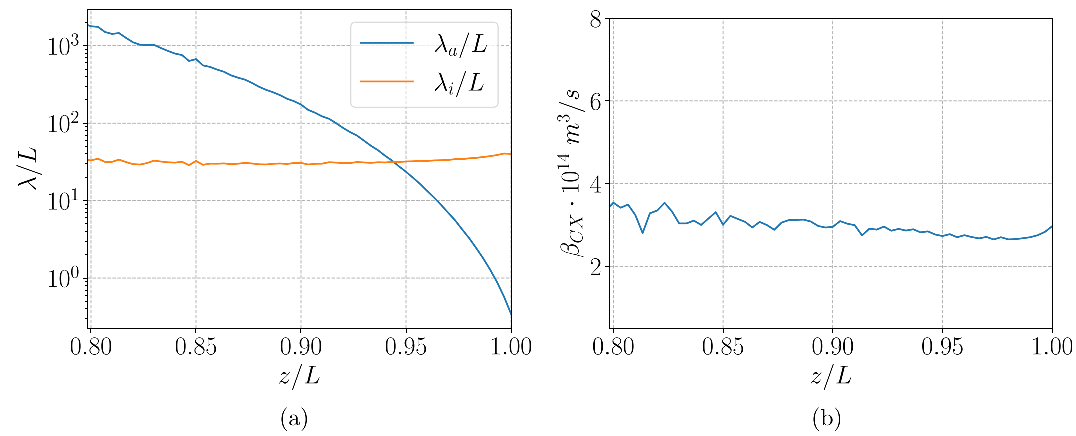

FIG. 10. (a) Knudsen number parameter for atoms and ions and (b) collision rate due to CX. Note that the ionization collision rate is fixed to $\beta_i = 2.1 \times 10^{-14}$ m^3/s due to isothermal electrons.

the source term, n is the plasma density, and n_a is the atom density. Here, the source term S_{CX} is computed numerically as the weighted number of CX collisions occurred during the collision time step, divided by the cell volume and the collision time step.

The mean free path for ions is found similarly

$$\lambda_i = \frac{v_i}{n_a\langle\sigma v\rangle_{cx}} \tag{14}$$

with the ion flow velocity v_i and atom density n_a. Both λ_a and λ_i are depicted in Fig. 10(a) in the near-wall region for Base case. As expected, λ_a is smaller, although the flow regime for both species in the near-wall region is similar, the so-called transitional flow which is defined by Knudsen numbers between 0.1 and 10 (non-dimensional parameter defined as λ/L). The resulting collision rate for CX collisions is about 3×10^{-14} m^3/s in the near-wall region, which is $\sim 50\%$ higher than the ionization rate, Fig. 10(b).

The three components of total neutral atom yield are reflected atoms (most energetic), Franck–Condon atoms, and cold atoms of wall temperature. Due to a shorter mean free path cold atoms accumulate, and they are responsible for high atom density near the wall. This case results in a stationary plasma flow, and the atom phase space in equilibrium is shown in Fig. 11(b). CX reaction neutralize some of the accelerated ions and effectively creates a fast atom component bombarding the wall. For ions, in addition to fast ions accelerated in the mirror, high-density slow ion sources appear due to CX and ionization; see Fig. 11(a) for the ion distribution function (IDF) and the ADF. The slice of the IDF and the ADF in a close vicinity to the wall is shown in Fig. 12, where the main ion and atom populations are depicted.

B. Ion temperature effect

It was suggested previously[1] that an increase of the ion temperature reduces the atom density near the recycling wall. Here, we present

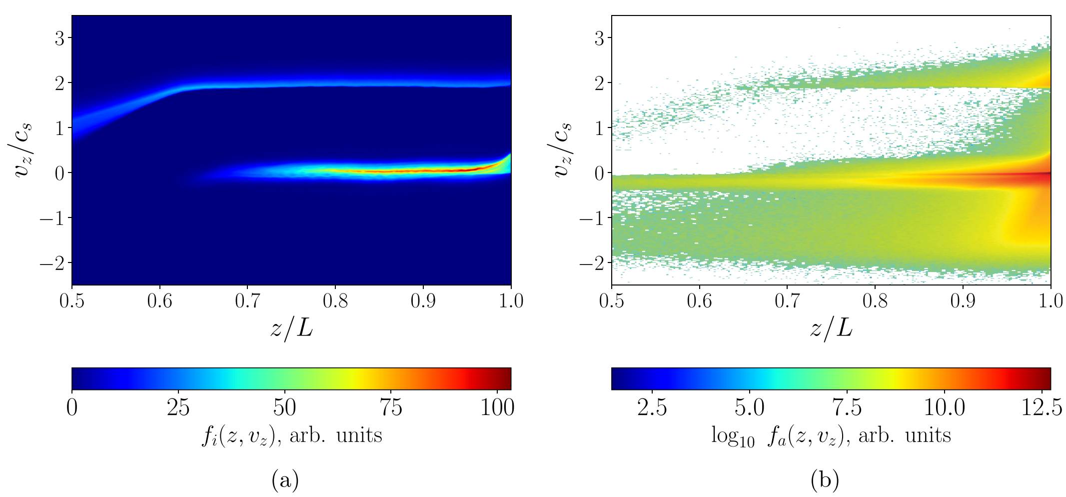

FIG. 11. Instantaneous plots of the distribution functions for (a) ions and (b) atoms in (z, v_z) space. Atoms are depicted in logarithmic scale due to its large density scale separation for various atom populations.

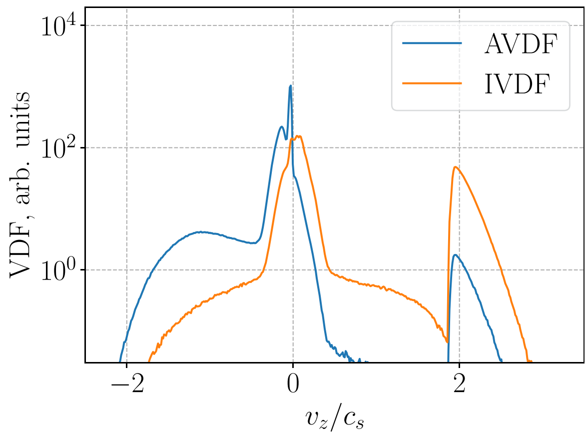

FIG. 12. Instantaneous plot of velocity distribution functions for ions and atoms close to the wall (at $z = 0.95L$). Ions accelerated to supersonic velocities bombarding the wall, with a notable fraction of them neutralized due to CX. The main component of neutral atom yield is wall-temperature ($T_{a,c}$) atoms, its density few orders higher than that of the reflected component. The ionization process leads to the generation of a relatively high-density ion population with low mean energy.

results for various values of the injected ion temperatures $T_i = (50, 100, 200)$ eV, keeping other parameters the same as in Base case, Table II. It is noted that the increased ion temperature slightly increases the ion density near the source region and reducing the ratio of escaped particles reaching the wall. This leads to a lower neutral atom yield near the wall. Figures 13(a)–13(d) show the resulted profiles for macroscopic variables. The neutral atom density decreases with the increase of ion temperature; this results into a weaker plasma source near the wall. As a result, the electrostatic potential drop across the whole domain is larger for a higher temperature. The corresponding mean free paths in the near-wall region are shown in Figs. 14(a) and 14(b), suggesting weaker collisionality for higher ion temperatures in the source.

C. Instability induced by charged-exchange interactions with cold atoms

Here, we present the cases resulting in ion–ion streaming instabilities in the near-the-wall region. The counterstreaming ion configuration is created by the accelerated original ion flow and the cold ion source (due to CX and ionization) near the wall. The density of the cold ion source is proportional to the atom density near the wall,

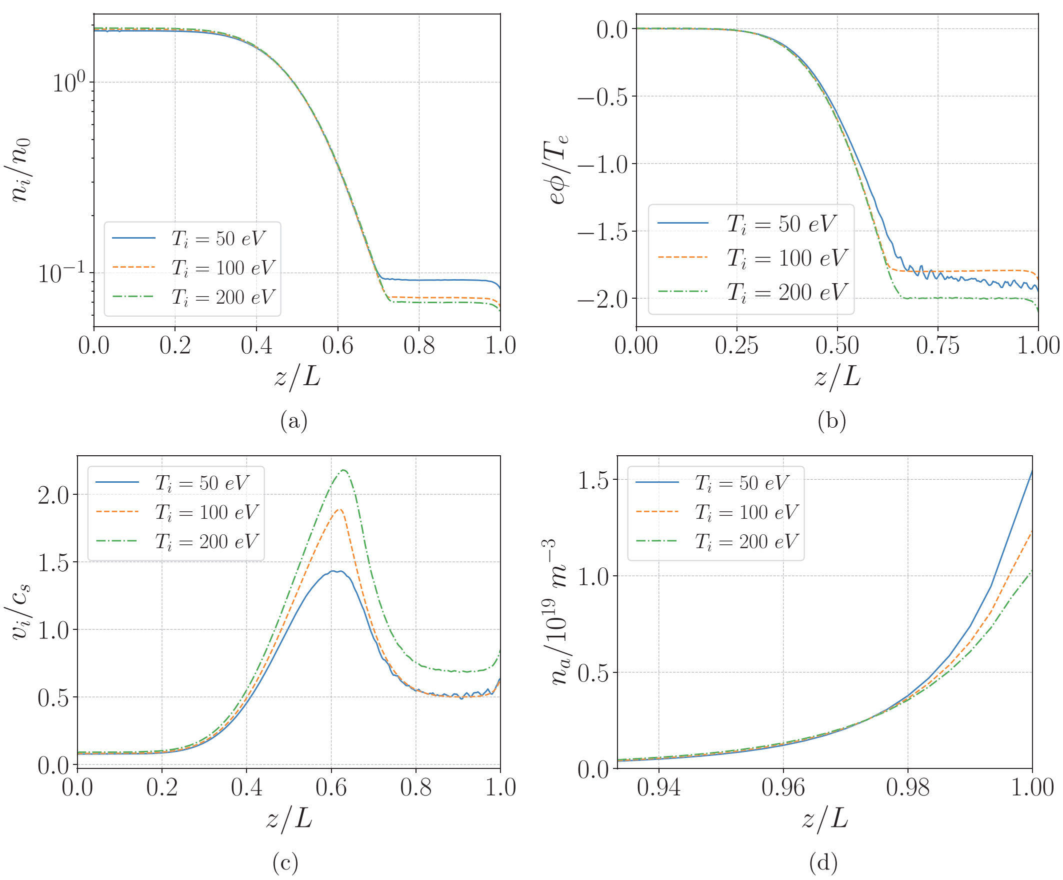

FIG. 13. (a) Spatial profiles of plasma density, (b) electrostatic potential, (c) ion velocity, and (d) atom density for varying ion temperature of the source. Note that atom density profile is shown in the near-wall region as its value drops significantly into the channel and thus collision effects become negligible.

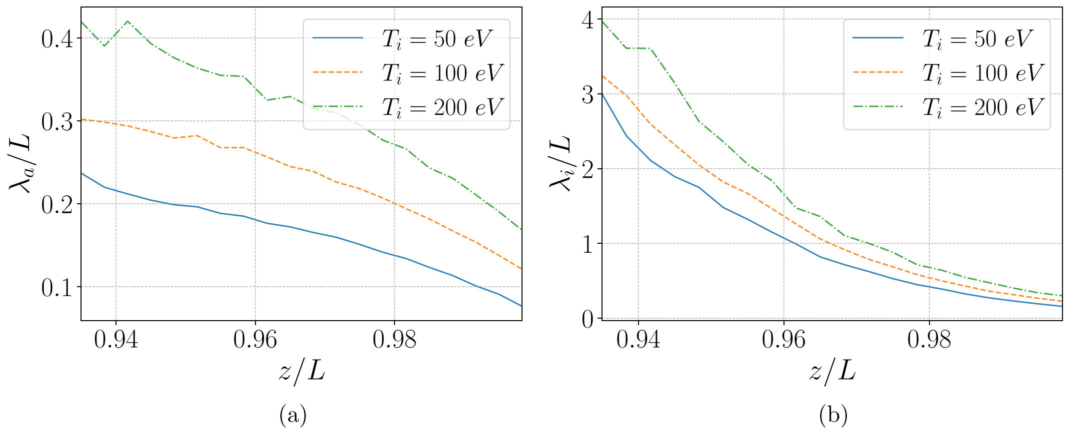

FIG. 14. Knudsen number for (a) atoms and (b) ions near the wall for varying ion temperature in the source.

which, in turn, is higher for the cold atom component, which we will control via the fraction of the Franck–Condon atoms P_{FC}. We present three cases, where the parameter is varied as $P_{FC} = (0.2, 0.5, 0.8)$, while all other parameters are kept the same as in Base case. The instability threshold is observed for the value $P_{FC} \approx 0.5$, and results in stronger turbulence with lower values of P_{FC}, since it increases the density of cold atom component.

We present the corresponding instantaneous IDF and ADF are shown in Figs. 15(a) and 15(b). For the reduced cold atom component with $P_{FC} = 0.8$, the overall flow is stable and close to equilibrium, see IDF and ADF plots in Figs. 16(a) and 16(b). Moreover, the oscillation amplitude of the global electrostatic potential is shown in Figs. 17(a) and 17(b) for $P_{FC} = 0.2$ and $P_{FC} = 0.5$. One can see that the highly turbulent regime reaches the amplitudes of an order of the electron temperature which may have a significant effect on the plasma flow and its transport. It is worth pointing out that the exact quantity of the cold atoms leading to the instability is hard to predict in experiments.

Estimates of the growth rate can be obtained with a linear dispersion relation, derived for two cold ion beams of various densities n_{01}, n_{02} moving relative to each other with speed v_0 is given by

$$1 - \frac{(1-\alpha)k^2 c_s^2}{(\omega - k v_0)^2} - \frac{\alpha k^2 c_s^2}{\omega^2} = 0 \tag{15}$$

with the full quasineutrality $n = n_e = n_{01} + n_{02}$, where the density fraction $\alpha = n_{02}/(n_{01} + n_{02})$ represents the beam, k is the wavenumber, and $c_s^2 = T_e/m_i$ is the ion sound velocity. The dispersion relation (15) is derived assuming zero temperatures of two beams, thus accounting for the first two fluid moments and the electron Boltzmann relation that couples densities to the electric field. The solutions to the dispersion relation (15) are shown for the mode number $kL/2\pi = 10$ (which roughly corresponds to the observed unstable wavenumber), with $L = 1$ m, in Fig. 18. The maximal growth rate is observed for $\alpha = 0.5$, i.e., beams of equal densities, and the relative

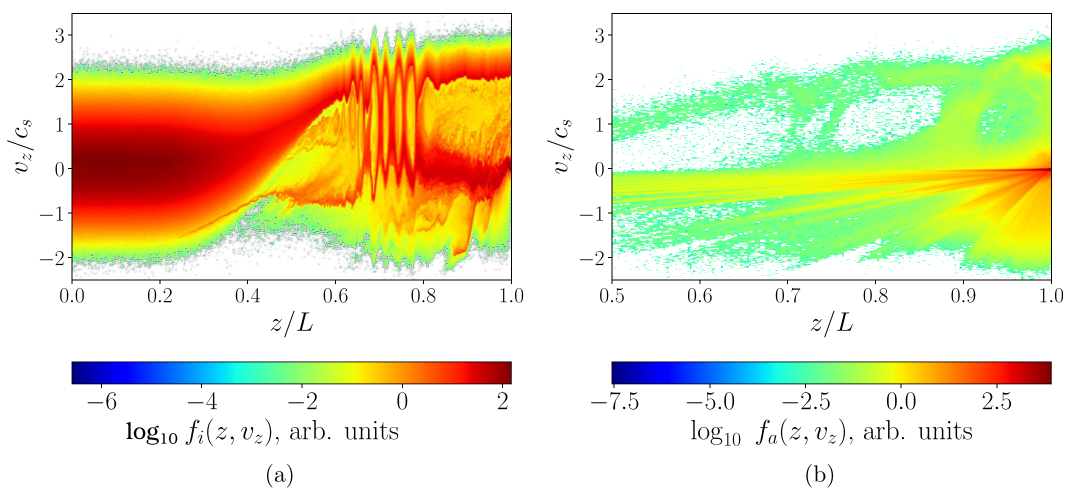

FIG. 15. Instantaneous plots of the distribution functions for (a) ions and (b) atoms in (z, v_z)-space, case for low fraction of Franck–Condon atoms, $P_{FC} = 0.2$, resulting in ion–ion streaming instability.

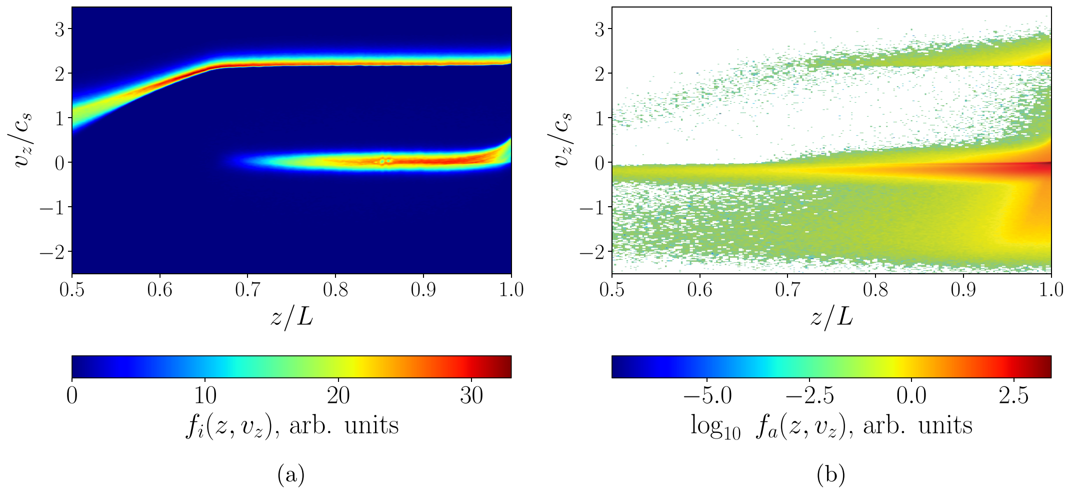

FIG. 16. Instantaneous plots of the distribution functions for (a) ions and (b) atoms in (z, v_z)-space, for a high fraction of Franck–Condon atoms, $P_{FC} = 0.8$, resulting in stable configuration.

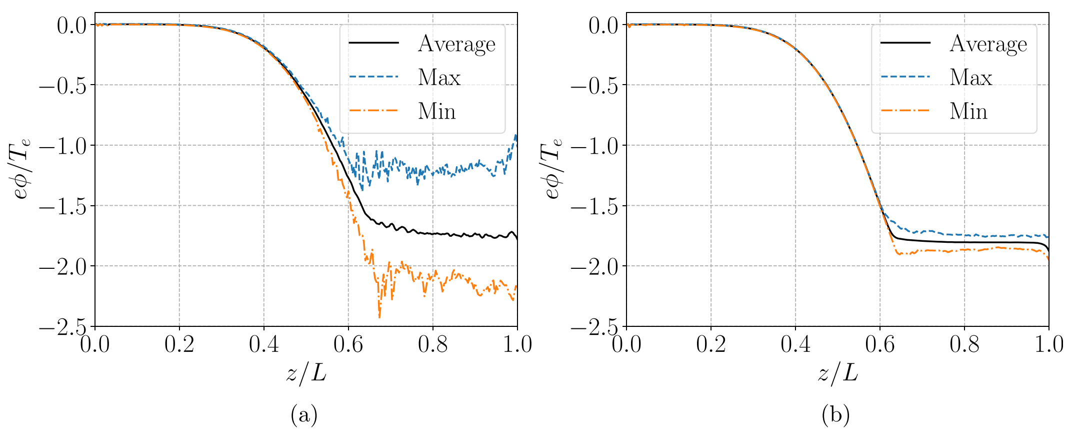

FIG. 17. Averaged macroscopic profiles along with maximum and minimum temporal values for the electrostatic potential, for (a) oscillatory case with $P_{FC} = 0.2$ and (b) near-stable case with $P_{FC} = 0.5$. The behavior of plasma density and ion velocity profiles is similar, resulting the relatively small difference of averaged profiles for oscillatory and stable cases (not shown here).

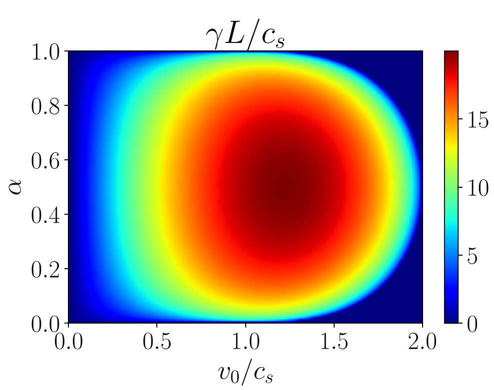

FIG. 18. Growth rate $\gamma = \mathrm{Im}(\omega)$, obtained by solving the fluid dispersion Eq. (15), as a function of the relative velocity between ion beams and the fraction of stationary beam density.

beam velocity $v_0 \approx 1.3c_s$. Note that the dispersion equation derived here assumes quasineutrality and neglects ion temperature effects with the goal to obtain approximate values of the growth rates. The resulted growth rates are normalized to the ion bounce period in a given system, and the obtained values suggest that the instability has sufficient time to develop during the simulations.

The files are named `nozzle_all_atoms_pfc0d.mp4` to indicate the value of P_{FC}, with `d` representing the tenth decimal place. In the top row from left to right it shows ion phase space, atom phase space, and macroscopic density profiles. In the bottom row from left to right it shows ion velocity, plasma potential, and velocity distribution function slices evaluated at the wall ($z = L$).

IV. SUMMARY AND CONCLUSIONS

Divertors in linear fusion devices confine plasma and control output power, but plasma–wall contact is unavoidable.[3,29] Plasma–wall interactions lead to numerous effects, and one of the most important is the recycling. A population of neutral atoms near the wall leads to additional ionization and plasma sources that affect the plasma flow by decreasing the overall potential drop and even may lead to the excitation of instabilities. We have developed a hybrid model for large-scale plasma flow in a magnetic nozzle incorporating drift-kinetic ions, kinetic atoms, and a Boltzmann electron model, as well as a plasma–wall recycling model. The population of neutral atoms consists of energetic particles due to reflection and low-energy atoms due to surface desorption. Atom collisions, CX with ions, and ionization by electron impact were included.

The results presented in this work show that the desorbed atom population may significantly impact the global plasma potential, creating high-density cold plasma sources near the wall, which restrict ion penetration and potentially increase electron losses. It is shown that with increasing of the ion temperature, the ion recycling decreases, resulting in the lower density of plasma sources near the wall.

We have identified the ion–ion streaming instability that can be triggered by the high-density cold plasma source in the near-the-wall region. The instability can be excited due to the presence of two groups of ions: the accelerated ions and the ions with low velocity formed in the near-wall region due to neutral ionization and CX. Nevertheless, even with present fluctuations, the averaged plasma profiles do not change significantly. Such fluctuations may have a stronger effect on electron transport which is not included in our model. The identified instability may be a source of background pressure facility effects observed in some experiments with accelerated ion beams.[30,31] Our work is mostly motivated by the mirror fusion applications;[1,3] however, some results may be relevant to plasma propulsion.[32,33] It has been suggested that the interaction of the accelerated ion beam with neutrals may lead to enhanced thrust.[34] These effects can also be studied with our model; however, such studies are left for future work.

SUPPLEMENTARY MATERIAL

See the supplementary material for time-dependent evolution of the main plasma parameters for the experiments in Sec. III C, by including the video files supplied with this manuscript. The files names `nozzle_all_atoms_pfc0d.mp4` indicate the value of P_{FC}, with `d` representing the tenth decimal place. In the top row, it depicts the evolution of ion phase space (top left), atom phase space (top center), and macroscopic density profiles (top right). In the bottom row: ion velocity (bottom left), plasma potential (bottom center), and velocity distribution function slices evaluated at the wall ($z = L$) (bottom right).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Oleksandr Chapurin: Conceptualization (equal); Data curation (lead); Formal analysis (equal); Investigation (lead); Methodology (equal); Project administration (supporting); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Marilyn Jimenez:** Investigation (supporting); Software (equal); Visualization (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Andrei I. Smolyakov:** Conceptualization (lead); Funding acquisition (equal); Methodology (equal); Project administration (lead); Supervision (lead); Validation (equal); Writing – original draft (supporting); Writing – review & editing (equal). **Peter Yushmanov:** Conceptualization (equal); Funding acquisition (equal); Methodology (equal); Supervision (equal); Validation (supporting); Writing – review & editing (supporting). **Sean Dettrick:** Conceptualization (equal); Funding acquisition (supporting); Methodology (equal); Validation (supporting); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1] D. Ryutov, P. Yushmanov, D. Barnes, and S. Putvinski, "Divertor for a linear fusion device," AIP Conf. Proc. **1721**, 060003 (2016).
[2] D. Ryutov, "Axial electron heat loss from mirror devices revisited," Fusion Sci. Technol. **47**, 148–154 (2005).
[3] M. Binderbauer, T. Tajima, L. Steinhauer, E. Garate, M. Tuszewski, L. Schmitz, H. Guo, A. Smirnov, H. Gota, D. Barnes *et al.*, "A high performance field-reversed configuration," Phys. Plasmas **22**, 056110 (2015).
[4] S. Andrews, S. Di Fede, and M. Magarotto, "Fully kinetic model of plasma expansion in a magnetic nozzle," Plasma Sources Sci. Technol. **31**, 035022 (2022).
[5] Z. Y. Chen, Y. B. Wang, J. X. Ren, H. B. Tang, P. Wu, and M. Li, "The fully-kinetic investigations on the ion acceleration mechanisms in an electron-driven magnetic nozzle," Plasma Sources Sci. Technol. **31**, 055013 (2022).
[6] B. A. Wetherton, A. Le, J. Egedal, C. Forest, W. Daughton, A. Stanier, and S. Boldyrev, "A drift kinetic model for the expander region of a magnetic mirror," Phys. Plasmas **28**, 042510 (2021).
[7] F. H. Ebersohn, J. Sheehan, A. D. Gallimore, and J. V. Shebalin, "Kinetic simulation technique for plasma flow in strong external magnetic field," J. Comput. Phys. **351**, 358–375 (2017).
[8] J. Zhou, G. Sanchez-Arriaga, and E. Ahedo, "Time-dependent expansion of a weakly-collisional plasma beam in a paraxial magnetic nozzle," Plasma Sources Sci. Technol. **30**, 045009 (2021).
[9] E. Ahedo and M. Merino, "Two-dimensional supersonic plasma acceleration in a magnetic nozzle," Phys. Plasmas **17**, 073501 (2010).
[10] J. J. Ramos, M. Merino, and E. Ahedo, "Three dimensional fluid-kinetic model of a magnetically guided plasma jet," Phys. Plasmas **25**, 061206 (2018).
[11] S. D. Baalrud, T. Lafleur, R. W. Boswell, and C. Charles, "Particle-in-cell simulations of a current-free double layer," Phys. Plasmas **18**, 063502 (2011).
[12] M. Jimenez, A. I. Smolyakov, O. Chapurin, and P. Yushmanov, "Ion kinetic effects and instabilities in the plasma flow in the magnetic mirror," Phys. Plasmas **29**, 112117 (2022).
[13] K. Takahashi, C. Charles, and R. W. Boswell, "Wave-driven electron inward transport in a magnetic nozzle," Sci. Rep. **12**, 20137 (2022).
[14] A. Fruchtman, K. Takahashi, C. Charles, and R. W. Boswell, "A magnetic nozzle calculation of the force on a plasma," Phys. Plasmas **19**, 033507 (2012).
[15] E. Thomas, R. Janev, and J. Smith, "Scaling of particle reflection coefficients," Nucl. Instrum. Methods Phys. Res., Sect. B **69**, 427–436 (1992).
[16] A. Fruchtman, G. Makrinich, P. Chabert, and J. M. Rax, "Enhanced plasma transport due to neutral depletion," Phys. Rev. Lett. **95**, 115002 (2005).
[17] S.-W. Ng and A. B. Hassam, "Neutral penetration in centrifugally confined plasmas," Phys. Plasmas **14**, 102508 (2007).
[18] A. Fruchtman, "Neutral depletion in a collisionless plasma," IEEE Trans. Plasma Sci. **36**, 403–413 (2008).

[19]K. Cartwright, J. Verboncoeur, and C. K. Birdsall, "Loading and injection of Maxwellian distributions in particle simulations," J. Comput. Phys. **162**, 483–513 (2000).

[20]D. D. Ryutov, "Open-ended traps," Phys. Usp. **31**, 300–327 (1988).

[21]V. Vahedi and M. Surendra, "A Monte Carlo collision model for the particle-in-cell method: Applications to argon and oxygen discharges," Comput. Phys. Commun. **87**, 179–198 (1995).

[22]W. Eckstein and H. Verbeek, "Reflection of light ions from solids," in *Data Compendium for Plasma-Surface Interactions*, special issue of Nuclear Fusion Volume 24, issue S1 (IOP, 1984).

[23]Y. Xia, X. Xu, C. Tan, L. Mei, H. Yang, and X. Sun, "Reflection of light ion from solid surfaces studied by Monte Carlo simulation and transport theory," J. Appl. Phys. **69**, 439–446 (1991).

[24]D. Stotler and C. Karney, "Neutral gas transport modeling with DEGAS 2," Contrib. Plasma Phys. **34**, 392–397 (1994).

[25]O. Chapurin, "Flows and instabilities in low-temperature plasmas with ionization and charge-exchange processes," Ph.D. thesis (University of Saskatchewan, 2022).

[26]G. Hagelaar, "Modelling methods for low-temperature plasmas," Habilitation thesis [Université Toulouse III Paul Sabatier (UT3 Paul Sabatier), 2008].

[27]R. K. Janev, W. D. Langer, K. Evans, Jr., and D. E. Post, Jr., *Elementary Processes in Hydrogen-Helium Plasmas: Cross Sections and Reaction Rate Coefficients*, Atoms and Plasmas Vol. 4 (Springer-Verlag, 1987).

[28]W. Hargus and M. Cappelli, "Interior and exterior laser-induced fluorescence and plasma measurements within a Hall thruster," J. Propul. Power **18**, 159–168 (2002).

[29]A. Beklemishev, A. Anikeev, V. Astrelin, P. Bagryansky, A. Burdakov, V. Davydenko, D. Gavrilenko, A. Ivanov, I. Ivanov, M. Ivantsivsky *et al.*, "Novosibirsk project of gas-dynamic multiple-mirror trap," Fusion Sci. Technol. **63**, 46–51 (2013).

[30]B. Wachs and B. Jorns, "Background pressure effects on ion dynamics in a low-power magnetic nozzle thruster," Plasma Sources Sci. Technol. **29**, 045002 (2020).

[31]I. G. Mikellides, A. L. Ortega, V. H. Chaplin, and J. S. Snyder, "Facility pressure effects on a Hall thruster with an external cathode, II: Theoretical model of the thrust and the significance of azimuthal asymmetries in the cathode plasma," Plasma Sources Sci. Technol. **29**, 035010 (2020).

[32]V. Saini and R. Ganesh, "Double layer formation and thrust generation in an expanding plasma using 1D-3V PIC simulation," Phys. Plasmas **27**, 093505 (2020).

[33]V. Saini and R. Ganesh, "Numerical simulation of a bi-directional plasma thruster for space debris removal," J. Plasma Phys. **88**, 905880203 (2022).

[34]G. Makrinich and A. Fruchtman, "Enhancement of electric force by ion-neutral collisions," Appl. Phys. Lett. **95**, 181504 (2009).